Notes: Greenstone, Hornbeck, and Moretti (JPE, 2010)

Identifying Agglomeration Spillovers: Evidence from Winners and Losers of Large Plant Openings

Contents

1 Big picture

- **Question.** Do large plant openings create productivity spillovers for incumbent plants nearby? The paper asks whether firms become more productive when a large manufacturing plant opens in their county.
- **Core empirical challenge.** Firms do not choose locations randomly. Counties that attract large plants may already have better infrastructure, labor markets, suppliers, industrial composition, or future growth prospects. A simple comparison between counties with and without new plants would therefore confound spillovers with selection.
- **Empirical design.** The paper compares "winning" counties that attracted a large "Million Dollar Plant" (MDP) to "losing" counties that were runner-up locations in the same corporate site-selection competition.
- **Main outcome.** The main outcome is total factor productivity (TFP) of incumbent manufacturing plants, estimated from plant-level production data in the Annual Survey of Manufactures and Census of Manufactures.
- **Main result.** Five years after a large plant opening, incumbent plants in winning counties have approximately 12% higher TFP than comparable incumbent plants in losing counties. The effect is economically large and corresponds to roughly $429 million in additional annual output in the preferred specification.
- **Mechanism result.** Spillovers are larger when incumbent industries are economically close to the new plant, especially when industries share labor pools or technologies. There is less evidence that input-output flows are the main channel.
- **Spatial equilibrium result.** Winning counties experience increased local economic activity and higher quality-adjusted wages. Productivity gains therefore do not translate one-for-one into profits because local factor prices rise.
- **Economic takeaway.** Agglomeration is not just a correlation between productive firms and productive locations. The winner-loser comparison provides evidence that a large plant can causally raise nearby incumbent productivity, but the gains are partly capitalized into local input prices.

2 Incremental contribution of the paper

- **From co-location correlations to quasi-experimental spillovers.** Earlier agglomeration work often relied on cross-sectional correlations between productivity and local density. This paper uses runner-up plant-location contests to create a more credible counterfactual.
- **Direct productivity measurement.** Rather than inferring spillovers from employment growth or co-agglomeration patterns, the paper estimates plant-level TFP directly from production data.
- **A novel "winners versus losers" research design.** The runner-up counties are selected by profit-maximizing firms as close alternatives to the winning counties. This design is more credible than comparing winners to the average U.S. county.

- **Mechanism contribution.** The paper does not stop at the average effect. It asks whether spillovers are stronger along labor-market, technology, customer, or supplier linkages.
- **Spatial equilibrium contribution.** The paper measures not only productivity effects but also the wage response and entry response, connecting agglomeration spillovers to equilibrium factor-price adjustment.
- **Policy contribution.** The findings speak directly to local industrial subsidies. If spillovers are heterogeneous across locations and industries, location incentives can sometimes internalize externalities, but subsidy policy remains risky because effects vary substantially across cases.
- **Robustness contribution.** The paper addresses production-function concerns using alternative functional forms, input interactions, region-by-year controls, unweighted regressions, fixed cost-share TFP, proxy-control methods, and instrumental-variable style approaches.

3 Theory and economic framework

3.1 Why agglomeration might raise productivity

- **Labor-market pooling.** Larger local labor markets can improve worker-firm matching, reduce vacancy risk for firms, and reduce unemployment risk for workers.
- **Input and service access.** Firms located near other firms may obtain specialized suppliers, repair services, logistics, or other local intermediate inputs more quickly or cheaply.
- **Knowledge spillovers.** Firms may learn from nearby firms through worker mobility, supplier interactions, imitation, informal contacts, patent-related knowledge flows, or technology adoption.
- **Amenities and natural advantages.** Some clusters may arise because productive amenities or worker-valued amenities are located in specific places. The empirical design is meant to separate spillovers from these fixed or preexisting advantages.

Interpretation: The paper's empirical focus is on factor-neutral productivity spillovers: the opening of a large plant may shift the production function of incumbent plants upward, allowing more output from the same measured inputs.

3.2 Simple spatial-equilibrium model

Let incumbent firms produce a nationally traded good using labor L, capital K, and land T:

$$\max_{L,K,T} f(A, L, K, T) - wL - rK - qT,$$

where A is total factor productivity, w is the wage, r is the rental cost of capital, and q is the price of land. Agglomeration enters through

$$A = A(N),$$

where N is the number of active firms in the county. A positive factor-neutral agglomeration spillover means

$$\frac{\partial A}{\partial N} > 0.$$

When another firm enters, incumbent profits change through two opposing forces:

$$\frac{d\Pi^*}{dN} = \frac{\partial f}{\partial A}\frac{\partial A}{\partial N} - \frac{\partial w}{\partial N}L^* - \frac{\partial q}{\partial N}T^*.$$

- The first term is the productivity benefit: agglomeration raises output for a given set of inputs.
- The second and third terms are cost effects: the new entrant increases local demand for labor and land, bidding up local input prices.
- In the short run, spillovers can raise profits if the productivity effect exceeds the local cost increase.
- In long-run spatial equilibrium, entry and factor-price adjustment should dissipate excess profits.

3.3 Empirical predictions

The model delivers four predictions that guide the empirical analysis:

- **Prediction 1: TFP should rise.** If there are positive spillovers, incumbent plants in winning counties should experience higher TFP after the MDP opening.
- **Prediction 2: spillovers may be stronger for nearby industries in economic space.** Plants sharing labor pools, technologies, customers, or suppliers with the MDP should benefit more.
- **Prediction 3: activity should expand.** If productivity gains are large enough, winning counties should attract more establishments or output.
- **Prediction 4: local factor prices should rise.** Firms compete for locally supplied inputs, especially labor and land, so wages should increase in winning counties.

4 Data and measurement

4.1 Million Dollar Plant sample

- The plant-opening sample comes from *Site Selection*'s "Million Dollar Plants" articles.
- These articles describe major corporate site-selection competitions and report the winning county and one or two runner-up counties.
- The paper identifies 47 usable manufacturing MDP openings that can be matched to Census plant-level data.
- A "case" is a plant-location contest: one winner county and one or more loser counties associated with the same new plant.
- The final sample includes 47 winning counties and 73 losing counties.

Interpretation: The MDP plants are selected, unusually large plants. This is useful for detecting spillovers, but it also means the estimated effects should not be interpreted as the effect of a typical plant opening.

4.2 Plant-level data

- The paper uses the Annual Survey of Manufactures (ASM) and Census of Manufactures (CM).
- These data contain plant-level output, employment, labor hours, materials, capital stocks, industry, county, and plant identifiers.
- The unique plant identifier allows the authors to follow incumbent plants over time.
- The main sample keeps incumbent plants that are present before the MDP opening and excludes plants owned by the MDP firm.
- The balanced event window is $t = -7$ through $t = 5$, where $t = 0$ is the MDP opening year.

4.3 Economic-distance measures

The paper measures proximity between incumbent industries and MDP industries using six measures:

- **CPS worker transitions:** worker flows across industries, meant to capture shared labor pools.
- **Citation pattern:** how often patents in one industry cite patents in the MDP industry.
- **Technology input:** R&D flows from the MDP industry to the incumbent industry.
- **Technology output:** R&D flows from the incumbent industry to the MDP industry.
- **Manufacturing input:** incumbent industry inputs purchased from the MDP industry.
- **Manufacturing output:** incumbent industry outputs purchased by the MDP industry.

Interpretation: The six measures allow the paper to distinguish several possible sources of agglomeration: labor pooling, technology or knowledge flows, and supplier-customer relationships.

5 Econometric model and empirical specifications

5.1 Augmented production function

The baseline production function is Cobb-Douglas:

$$Y_{pijt} = A_{pijt} L_{pijt}^{\beta_1} (K_{pijt}^B)^{\beta_2} (K_{pijt}^E)^{\beta_3} M_{pijt}^{\beta_4},$$

where p indexes plant, i industry, j case, and t event year. The inputs are labor hours, building capital, machinery and equipment capital, and materials.

Taking logs gives the main production-function regression:

$$\begin{aligned}\log Y_{pijt} = {} & \beta_1 \log L_{pijt} + \beta_2 \log K_{pijt}^B + \beta_3 \log K_{pijt}^E + \beta_4 \log M_{pijt} \\ & + \delta\, Winner_{pj} + \omega Trend_{jt} + \Theta(Trend_{jt} \times Winner_{pj}) \\ & + \kappa Post_{jt} + \gamma(Trend_{jt} \times Post_{jt}) \\ & + \nu_1(Winner_{pj} \times Post_{jt}) \\ & + \nu_2(Trend_{jt} \times Winner_{pj} \times Post_{jt}) \\ & + \alpha_p + \lambda_{it} + \mu_j + \varepsilon_{pijt}.\end{aligned}$$

- $Winner_{pj}$ equals one for incumbent plants located in the winning county.
- $Post_{jt}$ equals one after the MDP opening.
- α_p are plant fixed effects.
- λ_{it} are two-digit industry-by-year fixed effects.
- μ_j are case fixed effects.
- The main spillover parameters are ν_1 and ν_2.

5.2 Model 1: mean-shift difference-in-differences

Model 1 imposes no differential trend break and estimates a post-opening mean shift:

$$\log Y_{pijt} = \text{inputs} + \alpha_p + \lambda_{it} + \mu_j + \nu_1(Winner_{pj} \times Post_{jt}) + \varepsilon_{pijt}.$$

Interpretation: Model 1 asks whether incumbent plants in winning counties have higher average TFP after the opening than incumbent plants in losing counties, conditional on plant fixed effects, case fixed effects, and industry-year shocks.

5.3 Model 2: level shift plus trend break

Model 2 allows the MDP opening to change both the level and the post-opening trend of TFP. The reported effect after five years is evaluated at $t = 5$:

$$\text{Effect after 5 years} = \nu_1 + 6\nu_2,$$

where the six post-opening event years run from $t = 0$ through $t = 5$.

Interpretation: Model 2 is preferred when the effect appears to grow over time. It is also useful because it estimates and reports pre-trend differences between winner and loser counties.

5.4 Economic-distance regression

To test whether spillovers are stronger for economically close industries, the paper estimates a version of Model 1 with interactions:

$$\log Y_{pijt} = \text{baseline terms} + \rho_3(Winner_{pj} \times Post_{jt} \times Proximity_{ij}) + \varepsilon_{pijt}.$$

- $Proximity_{ij}$ is one of the standardized economic-distance measures.
- $\rho_3 > 0$ means the spillover is larger for incumbent plants whose industries are economically closer to the MDP industry.

6 Identification and empirical design

6.1 What identifies the effect

- The key comparison is within plant-location contests: winning counties versus runner-up losing counties.

- The identifying assumption is that, absent the MDP opening, incumbent plants in winning and losing counties within the same case would have followed similar TFP trends after conditioning on controls.
- This assumption is more plausible than comparing winners to all other U.S. counties because losing counties were finalist locations selected by the same profit-maximizing firm.

6.2 Evidence supporting the design

- Winning counties look different from the average U.S. county, confirming that the average county is not a good counterfactual.
- Winning and losing counties are more similar to each other on pre-opening observables.
- Table 3 shows that many observable county and plant characteristics are balanced between winners and losers.
- Figure 1 and Table 4 show that pre-opening TFP trends were similar in winning and losing counties.

6.3 Main threats

- **Unobserved differential shocks.** Winners may have received positive shocks exactly when the MDP opened. The event-study pre-trends, case fixed effects, and robustness controls reduce but cannot fully eliminate this concern.
- **Production-function bias.** Inputs are endogenous and TFP is a residual. The paper uses multiple alternative production-function specifications and input-correction methods.
- **Unobserved input quality or capacity utilization.** If input quality improved in winners, estimated TFP could rise mechanically. The paper checks energy use, public investment, input demand, and other channels.
- **Output prices.** The dependent variable is output value rather than physical quantity. The paper argues that manufacturing outputs are traded beyond the county and finds no strong evidence that local price effects drive the result.
- **External validity.** MDPs are large, selected plants that attracted local bidding. The effect may be an upper bound for typical plant openings.

7 Tables and figures

7.1 Table 1: The Million Dollar Plant Sample

Read:

- The table reports the composition of the 47 matched manufacturing MDP openings.
- There are 31 cases with one loser county and 16 cases with two loser counties.
- MDP openings are concentrated between 1981 and 1993.

- The MDPs are large: five years after opening, average output is about \$453 million and hours of labor are about 2.986 million.

Economic message: The treatment is economically meaningful. These are large plants that could plausibly affect local labor markets, supplier networks, and information flows.

7.2 Table 2: Summary Statistics for Measures of Industry Linkages

Read:

- The table defines six measures of economic proximity between incumbent industries and the MDP industry.
- CPS worker transitions capture shared labor pools.
- Patent citations and R&D flows capture technology proximity.
- Manufacturing input and output shares capture supplier-customer relationships.
- All measures are later normalized to mean zero and standard deviation one.

Economic message: The paper can test not only whether spillovers exist, but also whether they are larger when firms are connected through labor, technology, or input-output relationships.

TABLE 1
THE MILLION DOLLAR PLANT SAMPLE

	(1)
Sample MDP openings:[a]	
Across all industries	47
Within same two-digit SIC	16
Across all industries:	
Number of loser counties per winner county:	
1	31
2+	16
Reported year − matched year:[b]	
−2 to −1	20
0	15
1 to 3	12
Reported year of MDP location:	
1981–85	11
1986–89	18
1990–93	18
MDP characteristics, 5 years after opening:[c]	
Output (\$1,000s)	452,801
	(901,690)
Output, relative to county output 1 year prior	.086
	(.109)
Hours of labor (1,000s)	2,986
	(6,789)

[a] Million Dollar Plant openings that were matched to the Census data and for which there were incumbent plants in both winning and losing counties that are observed in each of the 8 years prior to the opening date (the opening date is defined as the earliest of the magazine reported year and the year observed in the SSEL). This sample is then restricted to include matches for which there were incumbent plants in the MDP's two-digit SIC in both locations.

[b] Only a few of these differences are 3. Census confidentiality rules prevent our being more specific.

[c] Of the original 47 cases, these statistics represent 28 cases. A few very large outlier plants were dropped so that the mean would be more representative of the entire distribution (those dropped had output greater than half of their county's previous output and sometimes much more). Of the remaining cases, most SSEL matches were found in the ASM or CM but not exactly 5 years after the opening date; a couple of SSEL matches in the 2xxx–3xxx SICs were never found in the ASM or CM; and a couple of SSEL matches not found were in the 4xxx SICs. The MDP characteristics are similar for cases identifying the effect within same two-digit SIC. Standard deviations are reported in parentheses. All monetary amounts are in 2006 U.S. dollars.

Table 1: The Million Dollar Plant Sample

TABLE 2
SUMMARY STATISTICS FOR MEASURES OF INDUSTRY LINKAGES

		Mean			
Measure of Industry Linkage	Description	All Plants	Only 1st Quartile	Only 4th Quartile	Standard Deviation
Labor market pooling:					
CPS worker transitions	Proportion of workers leaving a job in this industry that move to the MDP industry (15 months later)	.119	.002	.317	.249
Intellectual or technology spillovers:					
Citation pattern	Percentage of manufactured industry patents that cite patents manufactured in MDP industry	.022	.001	.057	.033
Technology input	R&D flows from MDP industry, as a percentage of all private-sector technological expenditures	.022	.000	.106	.084
Technology output	R&D flows to MDP industry, as a percentage of all original research expenditures	.011	.000	.042	.035
Proximity to customers and suppliers:					
Manufacturing input	Industry inputs from MDP industry, as a percentage of its manufacturing inputs	.017	.000	.075	.061
Manufacturing output	Industry output used by MDP industry, as a percentage of its output to manufacturers	.042	.000	.163	.139

NOTE.—The variable "CPS worker transitions" was calculated from the frequency of worker industry movements in the rotating CPS survey groups. This variation is by Census industry codes, matched to two-digit SIC values. The five other measures of cross-industry relationships were provided by Ellison et al. (forthcoming). These measures are defined in a three-digit SIC by three-digit SIC matrix, though much of the variation is at the two-digit level. In all cases, more positive values indicate a closer relationship between industries. Column 1 reports the mean value of the measure for all incumbent plants matched to their respective MDP. Column 2 reports the mean for the lowest 25 percent, and col. 3 reports the mean for the highest 25 percent. Column 4 reports the standard deviation across all observations. The sample of plants is all incumbent plants, as described for table 1, for which each industry linkage measure is available for the incumbent plant and its associated MDP. These statistics are calculated when weighting by the incumbent plant's total value of shipments 8 years prior to the MDP opening.

Table 2: Summary Statistics for Measures of Industry Linkages

7.3 Table 3: County and Plant Characteristics by Winner Status, 1 Year Prior to a Million Dollar Plant Opening

Read:

- The table compares winning counties, losing counties, and all U.S. counties before the MDP opening.
- Winning counties differ substantially from the average U.S. county.
- Winning and losing counties are much more similar to each other than winners and the national average.
- Plant-level characteristics are also relatively balanced between winners and losers.

Economic message: The table motivates the identification strategy. The average county is not a credible counterfactual, but the runner-up loser counties look much more comparable to winners before treatment.

7.4 Table 4: Incumbent Plant Productivity, Relative to the Year of an MDP Opening

Read:

- The table reports event-time TFP coefficients for incumbents in winning and losing counties.
- The omitted year is $t = -1$, the year before the MDP opening.
- Before the opening, differences between winners and losers are small and statistically indistinguishable.
- After the opening, the winner-minus-loser gap rises, reaching about 7.7 percentage points by year 5 in this event-study presentation.

Economic message: The table provides the core pre-trend and post-opening evidence. The identifying assumption is supported by flat pre-trends, while the post-opening divergence suggests a spillover effect.

TABLE 3

County and Plant Characteristics by Winner Status, 1 Year Prior to a Million Dollar Plant Opening

	All Plants					Within Same Industry (Two-Digit SIC)				
	Winning Counties (1)	Losing Counties (2)	All U.S. Counties (3)	t-Statistic (Col. 1 − Col. 2) (4)	t-Statistic (Col. 1 − Col. 3) (5)	Winning Counties (6)	Losing Counties (7)	All U.S. Counties (8)	t-Statistic (Col. 6 − Col. 7) (9)	t-Statistic (Col. 6 − Col. 8) (10)
	A. County Characteristics									
No. of counties	47	73				16	19			
Total per capita earnings ($)	17,418	20,628	11,259	**−2.05**	**5.79**	20,230	20,528	11,378	−.11	**4.62**
% change, over last 6 years	.074	.096	.037	−.81	1.67	.076	.089	.057	−.28	.57
Population	322,745	447,876	82,381	−1.61	**4.33**	357,955	504,342	83,430	−1.17	**3.26**
% change, over last 6 years	.102	.051	.036	**2.06**	**3.22**	.070	.032	.031	1.18	1.63
Employment-population ratio	.535	.579	.461	−1.41	**3.49**	.602	.569	.467	.64	**3.63**
Change, over last 6 years	.041	.047	.023	−.68	**2.54**	.045	.038	.028	.39	1.57
Manufacturing labor share	.314	.251	.252	**2.35**	**3.12**	.296	.227	.251	1.60	1.17
Change, over last 6 years	−.014	−.031	−.008	1.52	−.64	−.030	−.040	−.007	.87	**−3.17**
	B. Plant Characteristics									
No. of sample plants	18.8	25.6	7.98	−1.35	**3.02**	2.75	3.92	2.38	−1.14	.70
Output ($1,000s)	190,039	181,454	123,187	.25	**2.14**	217,950	178,958	132,571	.41	1.25
% change, over last 6 years	.082	.082	.118	.01	−.97	−.061	.177	.182	−1.23	**−3.38**
Hours of labor (1,000s)	1,508	1,168	877	1.52	**2.43**	1,738	1,198	1,050	.92	1.33
% change, over last 6 years	.122	.081	.115	.81	.14	.160	.023	.144	.85	.13

Note.—For each case to be weighted equally, counties are weighted by the inverse of their number per case. Similarly, plants are weighted by the inverse of their number per county multiplied by the inverse of the number of counties per case. The sample includes all plants reporting data in the ASM for each year between the MDP opening and 8 years prior. Excluded are all plants owned by the firm opening an MDP. Also excluded are all plants from two uncommon two-digit SIC values so that subsequently estimated clustered variance matrices would always be positive definite. The sample of all U.S. counties excludes winning counties and counties with no manufacturing plant reporting data in the ASM for 9 consecutive years. These other U.S. counties are given equal weight within years and are weighted across years to represent the years of MDP openings. Reported t-statistics are calculated from standard errors clustered at the county level. t-statistics greater than 2 are reported in bold. All monetary amounts are in 2006 U.S. dollars.

Table 3: County and Plant Characteristics by Winner Status, 1 Year Prior to a Million Dollar Plant Opening

TABLE 4

Incumbent Plant Productivity, Relative to the Year of an MDP Opening

Event Year	In Winning Counties (1)	In Losing Counties (2)	Difference Col. 1 − Col. 2 (3)
$\tau = -7$	.067	.040	.027
	(.058)	(.053)	(.032)
$\tau = -6$	.047	.028	.018
	(.044)	(.046)	(.023)
$\tau = -5$	.041	.021	.020
	(.036)	(.040)	(.025)
$\tau = -4$	−.003	.012	−.015
	(.030)	(.030)	(.024)
$\tau = -3$	.011	−.013	.024
	(.022)	(.022)	(.021)
$\tau = -2$	−.003	.001	−.005
	(.027)	(.011)	(.028)
$\tau = -1$	0	0	0
$\tau = 0$	.013	−.010	.023
	(.018)	(.011)	(.019)
$\tau = 1$	.023	−.028	.051**
	(.026)	(.024)	(.023)
$\tau = 2$	.004	−.046	.050
	(.036)	(.046)	(.033)
$\tau = 3$	.003	−.073	.076*
	(.047)	(.057)	(.043)
$\tau = 4$	.004	−.072	.076**
	(.053)	(.062)	(.033)
$\tau = 5$	−.023	−.100	.077**
	(.069)	(.067)	(.035)
R^2	.9861		
Observations	28,732		

Note.—Standard errors are clustered at the county level. Columns 1 and 2 report coefficients from the same regression: the natural log of output is regressed on the natural log of inputs (all worker hours, building capital, machinery capital, materials), year by two-digit SIC fixed effects, plant fixed effects, case fixed effects, and the reported dummy variables for whether the plant is in a winning or losing county in each year relative to the MDP opening. When a plant is a winner or loser more than once, it receives a dummy variable for each incident. Plant-year observations are weighted by the plant's total value of shipments 8 years prior to the MDP opening. Data on plants in all cases are available only 8 years prior to the MDP opening and 5 years after. Capital stocks were calculated using the permanent inventory method from early book values and subsequent investment. The sample of incumbent plants is the same as in cols. 1 and 2 of table 3.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 4: Incumbent Plant Productivity, Relative to the Year of an MDP Opening

7.5 Figure 1: All Incumbent Plants' Productivity in Winning versus Losing Counties, Relative to the Year of an MDP Opening

Read:

- The top panel plots TFP separately for winning and losing counties.
- The bottom panel plots the winner-minus-loser difference.
- Before the opening, winning and losing counties have similar TFP trends.
- After the opening, the difference rises sharply.

Economic message: This figure is the visual heart of the paper. It shows that without loser counties, a before-after comparison in winners alone would miss the effect; the treatment effect appears as a relative divergence between winners and runners-up.

7.6 Table 5: Changes in Incumbent Plant Productivity Following an MDP Opening

Read:

- Model 1 estimates a post-opening mean shift of roughly 4.8% in the preferred MDP-county sample.
- Model 2 estimates an effect after five years of roughly 12.0%.
- The pre-trend coefficient is small and statistically insignificant.
- A random-winner placebo design would incorrectly suggest negative or small effects.

Economic message: Table 5 turns the event-study pattern into the paper's main quantitative estimate: a large and statistically significant productivity spillover, worth about $429 million in annual output five years after the opening.

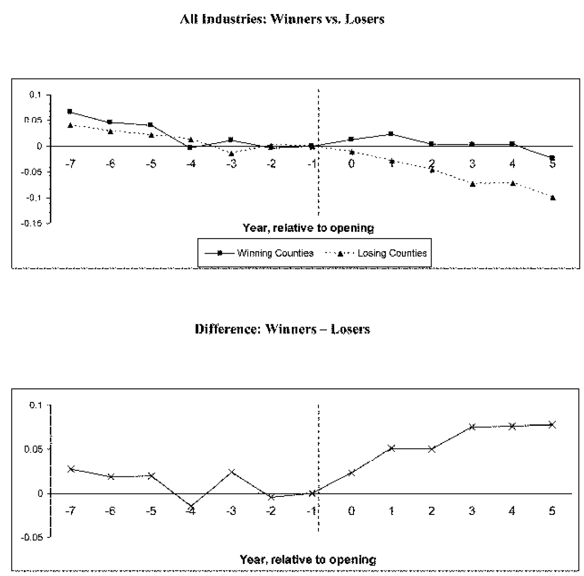

FIG. 1.—All incumbent plants' productivity in winning versus losing counties, relative to the year of an MDP opening. These figures accompany table 4.

TABLE 5

CHANGES IN INCUMBENT PLANT PRODUCTIVITY FOLLOWING AN MDP OPENING

	ALL COUNTIES: MDP WINNERS − MDP LOSERS		MDP COUNTIES: MDP WINNERS − MDP LOSERS		ALL COUNTIES: RANDOM WINNERS
	(1)	(2)	(3)	(4)	(5)
			A. Model 1		
Mean shift	.0442*	.0435*	.0524**	.0477**	− 0.0496***
	(.0233)	(.0235)	(.0225)	(.0231)	(.0174)
				[$170 m]	
R^2	.9811	.9812	.9812	.9860	~0.98
Observations (plant by year)	418,064	418,064	50,842	28,732	~400,000
			B. Model 2		
Effect after 5 years	.1301**	.1324**	.1355***	.1203**	−.0296
	(.0533)	(.0529)	(.0477)	(.0517)	(.0434)
				[$429 m]	
Level change	.0277	.0251	.0255	.0290	.0073
	(.0241)	(.0221)	(.0186)	(.0210)	(.0223)
Trend break	.0171*	.0179**	.0183**	.0152*	− 0.0062
	(.0091)	(.0088)	(.0078)	(.0079)	(.0063)
Pre-trend	−.0057	−.0058	−.0048	−.0044	−.0048
	(.0046)	(.0046)	(.0046)	(.0044)	(.0040)
R^2	.9811	.9812	.9813	.9861	~.98
Observations (plant by year)	418,064	418,064	50,842	28,732	~400,000
Plant and industry by year fixed effects	Yes	Yes	Yes	Yes	Yes
Case fixed effects	No	Yes	Yes	Yes	NA
Years included	All	All	All	$-7 \le \tau \le 5$	All

NOTE.—The table reports results from fitting several versions of eq. (8). Specifically, entries are from a regression of the natural log of output on the natural log of inputs, year by two-digit SIC fixed effects, plant fixed effects, and case fixed effects. In model 1, two additional dummy variables are included for whether the plant is in a winning county 7 to 1 years before the MDP opening or 0 to 5 years after. The reported mean shift indicates the difference in these two coefficients, i.e., the average change in TFP following the opening. In model 2, the same two dummy variables are included along with pre- and post-trend variables. The shift in level and trend are reported, along with the pre-trend and the total effect evaluated after 5 years. In cols. 1, 2, and 5, the sample is composed of all manufacturing plants in the ASM that report data for 14 consecutive years, excluding all plants owned by the MDP firm. In these models, additional control variables are included for the event years outside the range from $\tau = -7$ through $\tau = 5$ (i.e., −20 to −8 and 6 to 17). Column 2 adds the case fixed effects that equal one during the period that τ ranges from −7 through 5. In cols. 3 and 4, the sample is restricted to include only plants in counties that won or lost an MDP. This forces the industry by year fixed effects to be estimated solely from plants in these counties. For col. 4, the sample is restricted further to include only plant by year observations within the period of interest (where τ ranges from −7 to 5). This forces the industry by year fixed effects to be estimated solely on plant by year observations that identify the parameters of interest. In col. 5, a set of 47 plant openings in the entire country were randomly chosen from the ASM in the same years and industries as the MDP openings (this procedure was run 1,000 times, and reported are the means and standard deviations of those estimates). For all regressions, plant by year observations are weighted by the plant's total value of shipments 8 years prior to the opening. Plants not in a winning or losing county are weighted by their total value of shipments in that year. All plants from two uncommon two-digit SIC values were excluded so that estimated clustered variance-covariance matrices would always be positive definite. In brackets is the value in 2006 U.S. dollars from the estimated increase in productivity: the percentage increase is multiplied by the total value of output for the affected incumbent plants in the winning counties. Reported in parentheses are standard errors clustered at the county level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Figure 1: All Incumbent Plants' Productivity in Winning versus Losing Counties, Relative to the Year of an MDP Opening

Table 5: Changes in Incumbent Plant Productivity Following an MDP Opening

7.7 Figure 2: Distribution of Case-Specific Mean Shift Estimates, Following an MDP Opening

Read:

- The figure plots case-specific mean-shift estimates for the 47 MDP cases.
- The estimates are highly heterogeneous.

- Many cases show positive effects, but a nontrivial share show small or negative effects.
- The figure reports 45 estimates because two cases are excluded for Census confidentiality reasons.

Economic message: Agglomeration benefits are not automatic. The average effect is positive, but local governments face substantial uncertainty when bidding for plants.

7.8 Table 6: Changes in Incumbent Plant Output and Inputs Following an MDP Opening

Read:

- The table estimates effects on output and each input separately.
- In Model 1, output rises by about 12%, while labor, machinery, buildings, and materials rise by less or are imprecisely estimated.
- In Model 2, output still rises more than most inputs.
- These patterns are consistent with productivity growth rather than only input expansion.

Economic message: Table 6 provides a non-residual check on the TFP result. Incumbents appear to produce more without proportionally larger input increases.

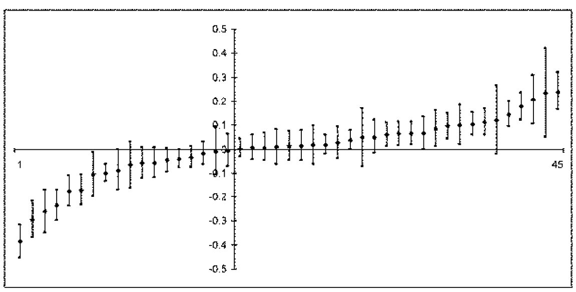

FIG. 2.—Distribution of case-specific mean shift estimates, following an MDP opening. The figure reports results from a version of model 1 that estimates the parameter θ_1 for each of the 47 MDP cases. The figure reports only 45 estimates because two cases were excluded for Census confidentiality reasons.

Figure 2: Distribution of Case-Specific Mean Shift Estimates, Following an MDP Opening

TABLE 6
CHANGES IN INCUMBENT PLANT OUTPUT AND INPUTS FOLLOWING AN MDP OPENING

	Output (1)	Worker Hours (2)	Machinery Capital (3)	Building Capital (4)	Materials (5)
Model 1: mean shift	.1200***	.0789**	.0401	.1327*	.0911***
	(.0354)	(.0357)	(.0348)	(.0691)	(.0302)
Model 2: after 5 years	.0826*	.0562	−.0089	−.0077	.0509
	(.0478)	(.0469)	(.0300)	(.0375)	(.0541)

NOTE.—The table reports results from fitting versions of eq. (8) for each of the indicated outcome variables (in logs). See the text for more details. Standard errors clustered at the county level are reported in parentheses.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

Table 6: Changes in Incumbent Plant Output and Inputs Following an MDP Opening

7.9 Table 7: Changes in Incumbent Plant Productivity Following an MDP Opening for Incumbent Plants in the MDP's Two-Digit Industry and All Other Industries

Read:

- The table separates incumbents in the MDP's own two-digit industry from incumbents in other industries.
- Model 1 estimates a 17% mean shift for same-industry incumbents, compared with 3.3% for other industries.
- Model 2 estimates are larger for the same industry but imprecise because only 16 cases have incumbents in the same two-digit industry.

Economic message: The spillover appears larger for economically closer firms. This is consistent with agglomeration mechanisms based on shared labor, technology, or learning.

7.10 Figure 3: Incumbent Plants' Productivity in the MDP's Two-Digit Industry, Winning versus Losing Counties, Relative to the Year of an MDP Opening

Read:

- Figure 3 shows the event-study pattern within the MDP's own two-digit industry.
- The estimates are noisy but the post-opening winner-minus-loser gap is large.
- There is no clear evidence of differential pre-trends before the opening.

Economic message: The most economically related plants appear to benefit most, but the sample is smaller and estimates are noisier.

TABLE 7
CHANGES IN INCUMBENT PLANT PRODUCTIVITY FOLLOWING AN MDP OPENING FOR INCUMBENT PLANTS IN THE MDP'S TWO-DIGIT INDUSTRY AND ALL OTHER INDUSTRIES

	All Industries (1)	MDP's Two-Digit Industry (2)	All Other Two-Digit Industries (3)
		A. Model 1	
Mean shift	.0477**	.1700**	.0326
	(.0231)	(.0743)	(.0253)
	[$170 m]	[$102 m]	[$104 m]
R^2	.9860	.9861	
Observations	28,732	28,732	
		B. Model 2	
Effect after 5 years	.1203**	.3289	.0889*
	(.0517)	(.2684)	(.0504)
	[$429 m]	[$197 m]	[$283 m]
Level change	.0290	.2814***	.0004
	(.0210)	(.0895)	(.0171)
Trend break	.0152*	.0079	.0147*
	(.0079)	(.0344)	(.0081)
Pre-trend	−.0044	−.0174	−.0026
	(.0044)	(.0265)	(.0036)
R^2	.9861	.9862	
Observations	28,732	28,732	

NOTE.—The table reports results from fitting versions of eq. (8). As a basis for comparison, col. 1 reports estimates from the baseline specification for incumbent plants in all industries (baseline estimates for incumbent plants in all industries, col. 4 of table 5). Columns 2 and 3 report estimates from a single regression, which fully interacts the winner/loser and pre/post variables with indicators for whether the incumbent plant is in the same two-digit industry as the MDP or a different industry. Reported in parentheses are standard errors clustered at the county level. The numbers in brackets are the value (2006 U.S. dollars) from the estimated increase in productivity: the percentage increase is multiplied by the total value of output for the affected incumbent plants in the winning counties.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

Table 7: Changes in Incumbent Plant Productivity Following an MDP Opening for Incumbent Plants in the MDP's Two-Digit Industry and All Other Industries

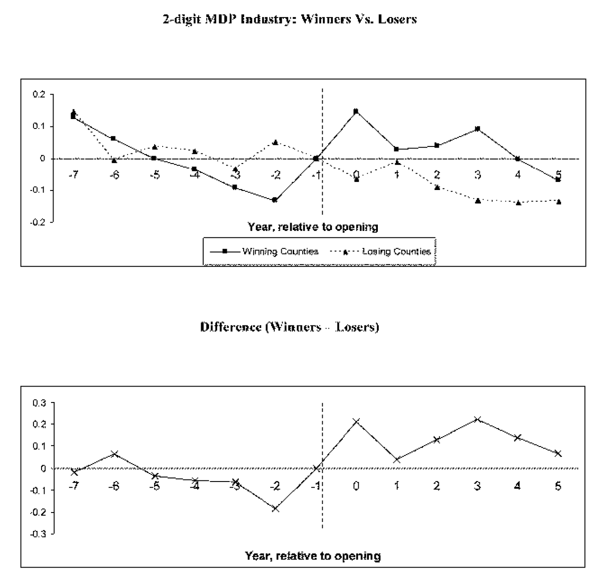

FIG. 3.—Incumbent plants' productivity in the MDP's two-digit industry, winning versus losing counties, relative to the year of an MDP opening. These figures accompany table 7, column 2 (MDP's two-digit industry).

Figure 3: Incumbent Plants' Productivity in the MDP's Two-Digit Industry, Winning versus Losing Counties, Relative to the Year of an MDP Opening

7.11 Figure 4: Incumbent Plants' Productivity in Other Industries (Not the MDP's Two-Digit Industry), Winning versus Losing Counties, Relative to the Year of an MDP Opening

Read:

- Figure 4 shows the event-study pattern for incumbents outside the MDP's two-digit industry.
- The post-opening divergence remains positive but is smaller than within the MDP industry.

- The figure supports the idea that spillovers decline with economic distance.

Economic message: Spillovers are not restricted to the MDP industry, but their magnitude is weaker for more distant industries.

7.12 Table 8: Changes in Incumbent Plant Productivity Following an MDP Opening, by Measures of Economic Distance between the MDP's Industry and Incumbent Plant's Industry

Read:

- A one-standard-deviation increase in CPS worker transitions is associated with a 7.0 percentage point larger spillover.
- Patent citation and technology-flow measures are also positive and statistically meaningful when entered separately.
- Manufacturing input and output linkages are small and statistically insignificant.
- When all measures enter jointly, estimates are less precise.

Economic message: The strongest evidence points toward labor-pool and technology channels, not supplier-customer flows. Spillovers seem to travel through people and knowledge more than through ordinary goods transactions.

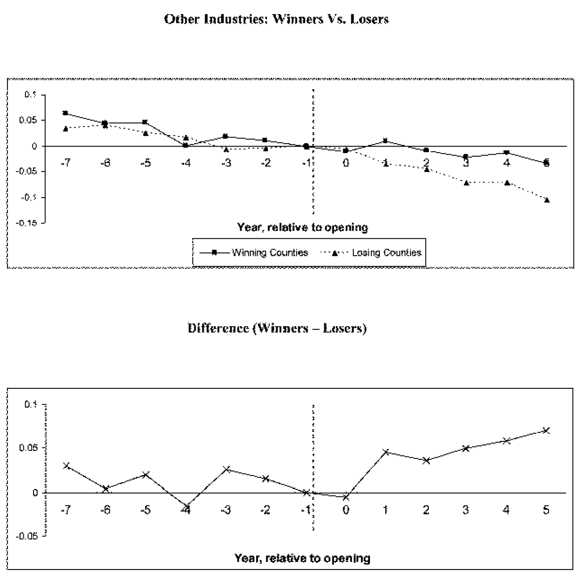

Fig. 4.—Incumbent plants' productivity in other industries (not the MDP's two-digit industry), winning versus losing counties, relative to the year of an MDP opening. These figures accompany table 7, column 3 (all two-digit industries, except the MDP's two-digit industry).

Figure 4: Incumbent Plants' Productivity in Other Industries (Not the MDP's Two-Digit Industry), Winning versus Losing Counties, Relative to the Year of an MDP Opening

TABLE 8
Changes in Incumbent Plant Productivity Following an MDP Opening, by Measures of Economic Distance between the MDP's Industry and Incumbent Plant's Industry

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CPS worker transitions	.0701***						.0374
	(.0237)						(.0260)
Citation pattern		.0545***					.0256
		(.0192)					(.0208)
Technology input			.0320*				.0501
			(.0173)				(.0421)
Technology output				.0596***			.0004
				(.0216)			(.0434)
Manufacturing input					.0060		−.0473
					(.0123)		(.0289)
Manufacturing output						.0150	−.0145
						(.0196)	(.0230)
R^2	.9852	.9852	.9851	.9852	.9851	.9852	.9853
Observations	23,397	23,397	23,397	23,397	23,397	23,397	23,397

Note.—The table reports results from fitting versions of eq. (9), which is modified from eq. (8). Building on the model 1 specification in col. 4 of table 5, each column adds interaction terms between winner/loser and pre/post status with the indicated measures of how an incumbent plant's industry is linked to its associated MDP's industry (a continuous version of results in table 7). These industry linkage measures are defined and described in table 2, and here the measures are normalized to have a mean of zero and a standard deviation of one. The sample of plants is that in col. 4 of table 5, but it is restricted to plants that have industry linkage data for each measure. For assigning this linkage measure, the incumbent plant's industry is held fixed at its industry the year prior to the MDP opening. Whenever a plant is a winner or loser more than once, it receives an additive dummy variable and interaction term for each occurrence. Reported in parentheses are standard errors clustered at the county level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 8: Changes in Incumbent Plant Productivity Following an MDP Opening, by Measures of Economic Distance between the MDP's Industry and Incumbent Plant's Industry

7.13 Table 9: Changes in Counties' Number of Plants, Total Output, and Skill-Adjusted Wages Following an MDP Opening

Read:

- The number of manufacturing establishments increases by about 12.6% in winning counties.
- Total manufacturing output increases by about 14.5%, though the estimate is imprecise.
- Skill-adjusted wages increase by about 2.7%.
- These results match the model's prediction that productivity spillovers attract activity and bid up local factor prices.

Economic message: The wage increase means that productivity gains are partly passed through to workers and local factor owners. Incumbent profits rise less than TFP.

7.14 Table 10: Changes in Incumbent Plant Productivity Following an MDP Opening, Robustness to Different Specifications

Read:

- The table checks sensitivity to translog production functions, input-by-industry interactions, input-by-winner/post interactions, region-year fixed effects, region-year-industry fixed effects, and unweighted regressions.
- Weighted estimates remain positive and broadly similar across specifications.
- Unweighted estimates are much smaller, suggesting that spillovers are concentrated among larger plants.

Economic message: The baseline finding is not an artifact of a particular production-function specification. However, the effect is economically concentrated in larger plants, which receive more weight in output-weighted estimates.

TABLE 9

Changes in Counties' Number of Plants, Total Output, and Skill-Adjusted Wages Following an MDP Opening

	A. Census of Manufactures		B. Census of Population
	Dependent Variable: Log(Plants) (1)	Dependent Variable: Log(Total Output) (2)	Dependent Variable: Log(Wage) (3)
Difference-in-difference	.1255**	.1454	.0268*
	(.0550)	(.0900)	(.0139)
R^2	.9984	.9931	.3623
Observations	209	209	1,057,999

Note.—The table reports results from fitting three regressions. In panel A, the dependent variables are the log of number of establishments and the log of total manufacturing output in the county, based on data from the Census of Manufactures. Controls include county, year, and case fixed effects. Reported are the county-level difference-in-difference estimates for receiving an MDP opening. Because data are available every 5 years, depending on the census year relative to the MDP opening, the sample years are defined to be 1–5 years before the MDP opening and 4–8 years after the MDP opening. Thus, each MDP opening is associated with one earlier date and one later date. The col. 1 model is weighted by the number of plants in the county in years −6 to −10, and the col. 2 model is weighted by the county's total manufacturing output in years −6 to −10. In panel B, the dependent variable is log wage and controls include dummies for age by year, age squared by year, education by year, sex by race by Hispanic by citizen, and case fixed effects. Reported is the county-level difference-in-difference estimate for receiving an MDP opening. Because data are available every 10 years, the sample years are defined to be 1–10 years before the MDP opening and 3–12 years after the MDP opening. As in panel A, each MDP opening is associated with one earlier date and one later date. The sample is restricted to individuals who worked more than 26 weeks in the previous year, usually work more than 20 hours per week, are not in school, are at work, and work for wages in the private sector. The number of observations reported refers to unique individuals: some Integrated Public Use Microdata Series county groups include more than one Federal Information Processing Standard (FIPS), so all individuals in a county group were matched to each potential FIPS. The same individual may then appear in more than one FIPS, and observations are weighted to give each unique individual the same weight. Reported in parentheses are standard errors clustered at the county level.

* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

Table 9: Changes in Counties' Number of Plants, Total Output, and Skill-Adjusted Wages Following an MDP Opening

TABLE 10

Changes in Incumbent Plant Productivity Following an MDP Opening, Robustness to Different Specifications

	Baseline Specification (1)	Translog Functional Form (2)	Input-Industry Interactions (3)	Input-Winner, Input-Post (4)	Region-Year Fixed Effects (5)	Region-Year-Industry Fixed Effects (6)	Unweighted (7)
Model 1: mean shift	.0477**	.0471**	.0406*	.0571**	.0442*	.0369*	.0146
	(.0231)	(.0226)	(.0220)	(.0245)	(.0230)	(.0215)	(.0107)
Model 2: after 5 years	.1203**	.1053*	.0977**	.1177**	.1176**	.0879**	.0065
	(.0517)	(.0535)	(.0487)	(.0538)	(.0520)	(.0442)	(.0281)

Note.—The table reports results from fitting several versions of eq. (8). Column 1 reports estimates from the baseline specification (col. 4 of table 5). Column 2 uses a translog functional form for inputs. Column 3 allows the effect of each input to differ by two-digit SIC value. Column 4 allows the effect of inputs to differ in winning/losing counties and before/after the MDP opening. Column 5 includes region (nine census divisions) by year fixed effects. Column 6 includes region by year by industry fixed effects. Column 7 reports on an unweighted version of eq. (8). Reported in parentheses are standard errors clustered at the county level.

* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

Table 10: Changes in Incumbent Plant Productivity Following an MDP Opening, Robustness to Different Specifications

7.15 Appendix Table A1: Changes in Incumbent Plant Productivity Following an MDP Opening (and Production Function Coefficients for Plant Inputs), Robustness to Specifications Adjusting for the Endogeneity of Plant Inputs, Part 1

Read:

- Appendix Table A1 reports robustness checks designed to address endogenous input choices.
- Part 1 contains the beginning of the table and Model 1 results.
- Alternative methods include fixing input coefficients at cost shares, controlling for investment-capital or materials-capital interactions, and using lagged input changes as instruments.

Economic message: The Model 1 effect remains positive across input-endogeneity corrections, suggesting that the mean-shift result is not driven mechanically by biased input coefficients.

7.16 Appendix Table A1: Changes in Incumbent Plant Productivity Following an MDP Opening (and Production Function Coefficients for Plant Inputs), Robustness to Specifications Adjusting for the Endogeneity of Plant Inputs, Part 2

Read:

- Part 2 contains Model 2 estimates and table notes.
- The five-year effects are generally positive across alternative corrections.
- Some specifications are less precise, but the pattern is consistent with the baseline finding.

Economic message: The appendix reinforces the main interpretation: the estimated productivity gains are not easily explained by standard production-function endogeneity concerns.

TABLE A1

Changes in Incumbent Plant Productivity Following an MDP Opening (and Production Function Coefficients for Plant Inputs), Robustness to Specifications Adjusting for the Endogeneity of Plant Inputs

	Baseline Specification (1)	Fixed Input Cost Shares: Plant Level (2)	Fixed Input Cost Shares: SIC-3 Level (3)	Investment-Capital Interactions (4)	Materials-Capital Interactions (5)	Materials-Capital and Materials-Labor Interactions (6)	IV Inputs, Lagged Change (7)
	A. Model 1						
Mean shift	.0477**	.0364	.0325	.0391*	.0399*	.0354*	.0613**
	(.0231)	(.0228)	(.0241)	(.0222)	(.0216)	(.0208)	(.0250)
Building capital	.0384**	. . .	. . .	. . .	. . .	. . .	.0205
	(.0152)						(.0731)
Equipment capital	.0771***	. . .	. . .	. . .	. . .	. . .	.0856**
	(.0218)						(.0364)
Labor	.2915***	. . .	. . .	.2660***	.2891***	. . .	.2197***
	(.0290)			(.0249)	(.0270)		(.0448)
Materials	.4504***	. . .	. . .	.4284***	. . .	. . .	.4359***
	(.0218)			(.0279)			(.0436)
Observations	28,732	28,732	28,732	18,856	28,732	28,732	23,992

Appendix Table A1: Robustness to Specifications Adjusting for the Endogeneity of Plant Inputs, Part 1

	B. Model 2						
After 5 years	.1203**	.0971	.0938*	.0839	.1004**	.0869*	.1135
	(.0517)	(.0656)	(.0538)	(.0527)	(.0487)	(.0480)	(.0695)
Building capital	.0393**	. . .	. . .	. . .	. . .	. . .	.0215
	(.0169)						(.0727)
Equipment capital	.0790***	. . .	. . .	. . .	. . .	. . .	.0899**
	(.0149)						(.0371)
Labor	.2925***	. . .	. . .	.2668***	.2903***	. . .	.2217***
	(.0284)			(.0245)	(.0262)		(.0448)
Materials	.4510***	. . .	. . .	.4300***	. . .	. . .	.4348***
	(.0217)			(.0280)			(.0438)
Observations	28,732	28,732	28,732	18,856	28,732	28,732	23,992

Note.—The table reports results from fitting several versions of eq. (8). Column 1 reports estimates from the baseline specification (col. 4 of table 5) and the estimated coefficients on each plant production input (logs). Column 2 reports estimates when fixing the coefficient on each plant's inputs to be its average cost share over the sample period, where per-period capital costs were calculated from capital rental rates using Bureau of Labor Statistics data. Column 3 reports estimates when fixing the coefficient on each plant's inputs to be the average cost share for all plants in its three-digit SIC level. Column 4 controls for a fourth-degree polynomial function of log capital and log investment and the interaction of both functions (separately for both types of capital; see Olley and Pakes 1996). Column 5 includes the same controls as col. 4 but replaces log investment with log materials (see Levinsohn and Petrin 2003). Column 6 adds interactions between log materials and log labor to the controls in col. 5 (see Ackerberg et al. 2006). Column 7 instruments for each input with the lagged change in input from 2 years prior to 1 year prior, dropping the first two years of data for each plant (see Blundell and Bond 1998). Reported in parentheses are standard errors clustered at the county level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Appendix Table A1: Robustness to Specifications Adjusting for the Endogeneity of Plant Inputs, Part 2

8 Short summary

- The paper studies whether large manufacturing plant openings create agglomeration spillovers for nearby incumbent plants.
- The design compares winning counties to runner-up losing counties in the same plant-location contests.
- Incumbent plants in winning counties have similar pre-opening TFP trends to those in losing counties.
- After the opening, incumbent plants in winning counties experience a large relative TFP increase.
- The preferred estimate is about 12% higher TFP after five years.
- Spillovers are stronger when incumbents share labor pools and technologies with the MDP industry.
- Winning counties also see more manufacturing establishments and higher wages.
- The paper therefore provides evidence for economically large agglomeration spillovers, but also shows that productivity gains are partly offset by higher local factor costs.

9 Conclusion

9.1 Core conclusion

The paper provides quasi-experimental evidence that the opening of a large manufacturing plant can raise the productivity of incumbent manufacturing plants in the same county. The runner-up-county research design is the key to interpreting the results causally: losing counties are not random places, but locations that were seriously considered by the same firm and narrowly lost.

9.2 Economic intuition

A large plant changes the local economic environment. It may bring specialized workers, suppliers, production methods, information, and credibility to a location. Nearby firms can benefit from these changes. At the same time, the large plant and subsequent entrants compete for local inputs. Wages and other local costs rise, so some productivity gains are transferred to workers and local factor owners.

9.3 Incremental contribution revisited

The paper advances the agglomeration literature by combining three ingredients: a credible counterfactual based on plant-location runners-up, direct plant-level productivity measurement, and mechanism tests based on economic proximity. This combination allows the authors to move beyond documenting that similar firms co-locate and toward estimating the productivity consequences of a large plant opening.

9.4 Policy implications

The results complicate the standard view of local subsidies. If all places receive the same spillover from a plant, subsidies are mostly a transfer and can be socially wasteful. But if spillovers are heterogeneous and depend on the match between the plant and local industrial structure, then location incentives can sometimes help a plant internalize local external benefits. The policy problem is that these benefits are uncertain and heterogeneous across cases.

9.5 Limitations

- The MDPs are large, selected, subsidized plants; the results may not generalize to ordinary plant openings.
- The design is quasi-experimental, not randomized. Residual unobserved shocks to winning counties remain possible.
- TFP is estimated from output value and measured inputs, so output price changes, input quality, and production-function issues must be considered carefully.
- Mechanism measures are proxies. The paper provides suggestive evidence for labor and technology channels, but it cannot fully distinguish knowledge spillovers from better worker-firm matching.

9.6 Takeaway

Large plants can create local productivity spillovers, especially for economically related firms. Agglomeration benefits are real in this setting, but they are uneven, partly capitalized into wages and local prices, and most relevant for policy when the match between a new plant and the existing local industrial base is strong.